Svelte for the Modern Web

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What is the language of the web?

```html
<head>
    <title>Example Domain</title>

    <meta charset="utf-8" />
    <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <style type="text/css">
    body {
        background-color: #f0f0f2;
        margin: 0;
        padding: 0;
        font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif;

    }
    div {
        width: 600px;
        margin: 5em auto;
        padding: 2em;
        background-color: #fdfdff;
        border-radius: 0.5em;
        box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
        color: #38488f;
        text-decoration: none;
    }
    @media (max-width: 700px) {
        div {
            margin: 0 auto;
            width: auto;
        }
    }
    </style>
</head>

<body>
<div>
    <h1>Example Domain</h1>
    <p>This domain is for use in illustrative examples in documents. You may use this
    domain in literature without prior coordination or asking for permission.</p>
    <p><a href="https://www.iana.org/domains/example">More information...</a></p>
</div>
</body>
</html>
```

Why use Javascript?

- Better user experience
- Applications instead of static documents
- Do things locally, can handle a poor connection (or none at all!)

A Simple Button

```html
<!DOCTYPE html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta http-equiv="X-UA-Compatible" content="IE=edge" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Document</title>
  </head>
  <body>
    <span class="num">0</span>
    <button class="add">Add</button>
    <button class="dec">decrease</button>
  </body>
</html>

<script>
  let numberOfProcess = 0;

  const add = document.querySelector(".add");
  const dec = document.querySelector(".dec");
  const num = document.querySelector(".num");

  const clickHandler = () => {
    numberOfProcess++;
    num.innerHTML = numberOfProcess;
  };
  const clickHandler2 = () => {
    numberOfProcess--;
    num.innerHTML = numberOfProcess;
  };

  add.addEventListener("click", clickHandler);
  dec.addEventListener("click", clickHandler2);
</script>
```

Not ideal

- Mixing ways of thinking
  - Mutating data
  - Displaying data
- A single buttons concerns are spread throughout
- Repetitious

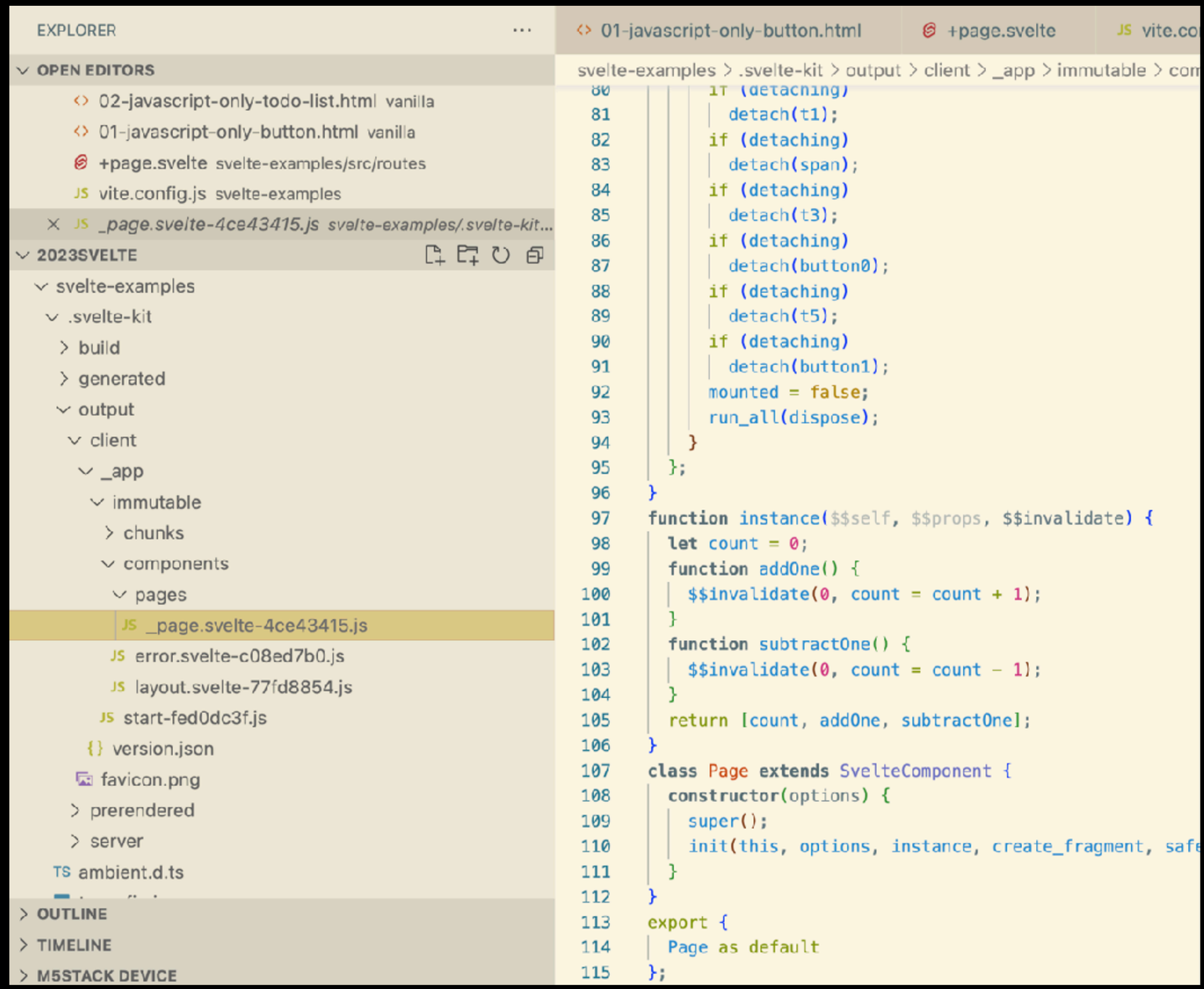
In Svelte

```
1   <script>
2     let count = 0;
3
4     function addOne() {
5       count = count + 1;
6     }
7     function subtractOne() {
8       count = count - 1;
9     }
10  </script>
11
12  <h1>Button Demo</h1>
13  <span>{count}</span>
14  <button on:click={addOne}>Increase</button>
15  <button on:click={subtractOne}>Decrease</button>
16
```

What is this wizardry?

How it works

- Svelte is compiled to Javascript
- Small bundle size
- Close to vanilla javascript performance
- Generated code on the right (normally minified)



Selections from the official documentation

https://svelte.dev/repl/ Japanese docs: https://svelte.jp/

Try these challenges in a REPL

- Basics
  - Make a self introduction, use variables in a template
- Reactivity
  - Make a button that when clicked shows the time
- Bindings
  - Temperature calculator

Final project

- More complex calculator
  - Distance, speed, time
- Countdown timer
- A web form with validation
- Your idea!

Extensions and extras

- Threlte
  - 3D library based on Three.js
- SvelteKit
  - Framework with routing, server side rendering
- Skeleton UI
  - UI library
- SvelteNative
  - Native apps for mobile using svelte

Basic outline

- What is the language of the web?
  - HTML
- Programming a basic component with javascript only
- Reactive svelte version of the same (in repl)
- Svelte project setup, deployment
- Reactivity, components, scoped styles, stores, animations
- Running calculator